

BLOCKCHAIN ENGINEERING

ASSIGNMENT - 010

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Batch: 05

Code -

```
Assignment-10.py > ...
1  from tkinter import *
2  from datetime import datetime
3
4  # Simulated Smart Contract
5  class SimpleStorageContract:
6      def __init__(self):
7          self.value = 0
8          self.transactions = []
9
10     def store_value(self, new_value):
11         self.value = new_value
12         tx_hash = f"0x{hash((new_value, datetime.now())):064x}"
13         self.transactions.append({
14             "value": new_value,
15             "time": datetime.now().strftime("%Y-%m-%d %H:%M:%S"),
16             "hash": tx_hash
17         })
18         return tx_hash
19
20     def get_value(self):
21         return self.value
22
23 # GUI Application
24 class SimpleStorageApp:
25     def __init__(self, root):
26         self.contract = SimpleStorageContract()
27         self.root = root
28         self.root.title("Simple Storage Contract Simulation")
29
30         self.entry = Entry(root, width=20)
31         self.entry.pack(pady=10)
32
33         self.store_button = Button(root, text="Store Value", command=self.store_value)
34         self.store_button.pack(pady=5)
35
36         self.status_label = Label(root, text="")
37         self.status_label.pack(pady=5)
38
39         self.hash_label = Label(root, text="")
40         self.hash_label.pack(pady=5)
41
42         self.history_label = Label(root, text="Transaction History:")
43         self.history_label.pack(pady=10)
44
45         self.history_text = Text(root, height=10, width=50)
46         self.history_text.pack()
```

